



# Lesson 4: Classification and Evaluation Metrics

Technologies for Artificial Intelligence

Fabio Antonini, Università degli Studi dell'Aquila

# Before we start

- Exercise 3 was due today
- Ridge, Lasso, and the choice of the penalty strength

# Today: from a number to a decision

- Logistic regression: the model and why the cost had to change
- The confusion matrix, and why accuracy lies
- Precision, recall, F1, and the threshold nobody chose
- Receiver operating characteristic (ROC) curves, and where they mislead
- Choosing a threshold from money

# The question has changed

- Lesson 3: given a house, **how many euros?**
- Lesson 4: given a disk drive, **will it fail?**

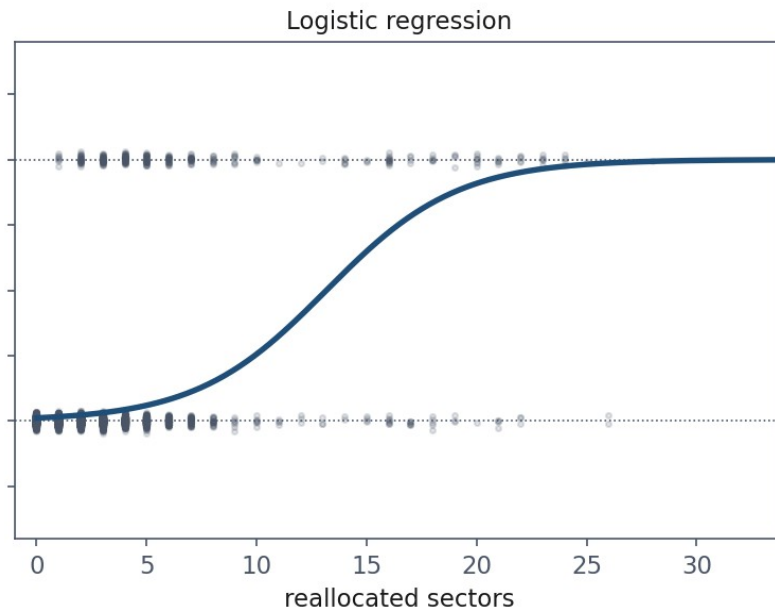
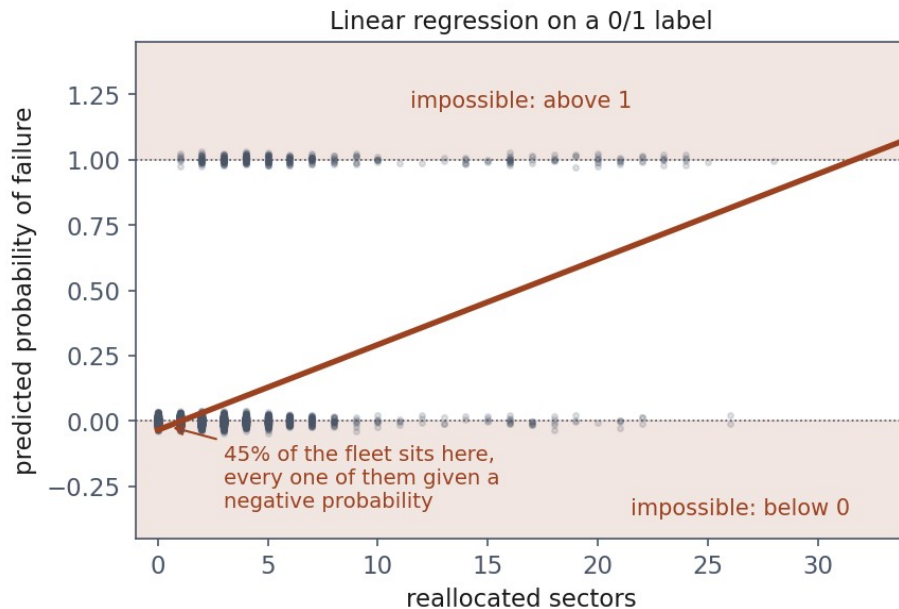
# The fleet

- **8,000 disk drives** in a data centre
- Six SMART counters each: Self-Monitoring, Analysis and Reporting Technology
- Label: did it fail within thirty days?

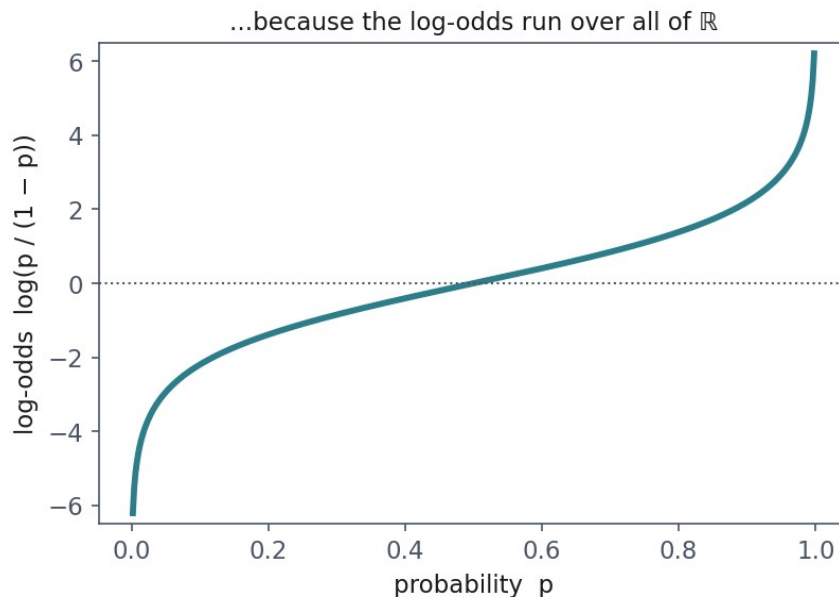
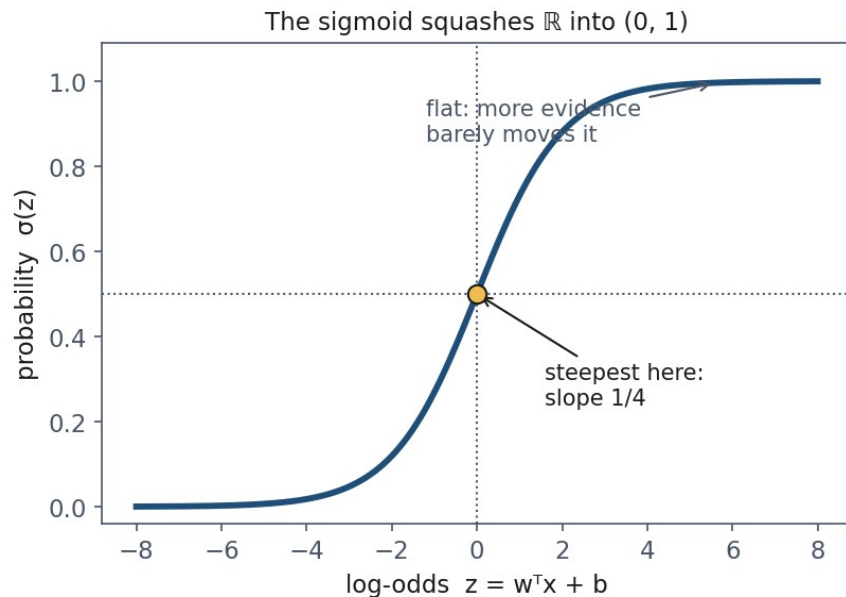
# 306

- **306 of the 8,000 drives failed.** 3.8% of the fleet
- Which means 96.2% did not

# What goes wrong if we just fit a line



# The sigmoid





# The model, in one line

$$P(y = 1 \mid x) = \sigma(w^\top x + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}}$$

# What the coefficients mean

- The model is linear in the log-odds, so  $\exp(\text{coefficient})$  multiplies the **odds**
- Features standardised, so one unit = one standard deviation
- A coefficient of 1.80 means **six times the odds** of failing

# The truth, and what we recovered

| Feature             | True        | Fitted      | Odds $\times$ |
|---------------------|-------------|-------------|---------------|
| reallocated_sectors | 1.80        | 1.66        | 5.28          |
| spin_retry_count    | 1.15        | 1.17        | 3.22          |
| read_error_rate     | 0.85        | 0.87        | 2.39          |
| temperature_c       | 0.45        | 0.47        | 1.59          |
| power_on_hours      | 0.35        | 0.38        | 1.46          |
| seek_error_rate     | <b>0.00</b> | <b>0.02</b> | <b>1.03</b>   |

# The intercept is the base rate

- Fitted intercept: **-6.09**
- $\sigma(-6.09) = 0.0023$ : a 0.2% chance for an average drive

# Fitting means minimising something, but what?

- Squared error worked for lesson 3
- Why not just apply it to the sigmoid's output?

# Maximum likelihood, in one sentence

- The model has a knob; each setting assigns a probability to what happened
- **Turn the knob so that what actually happened is least surprising**

# Cross-entropy, also called log loss

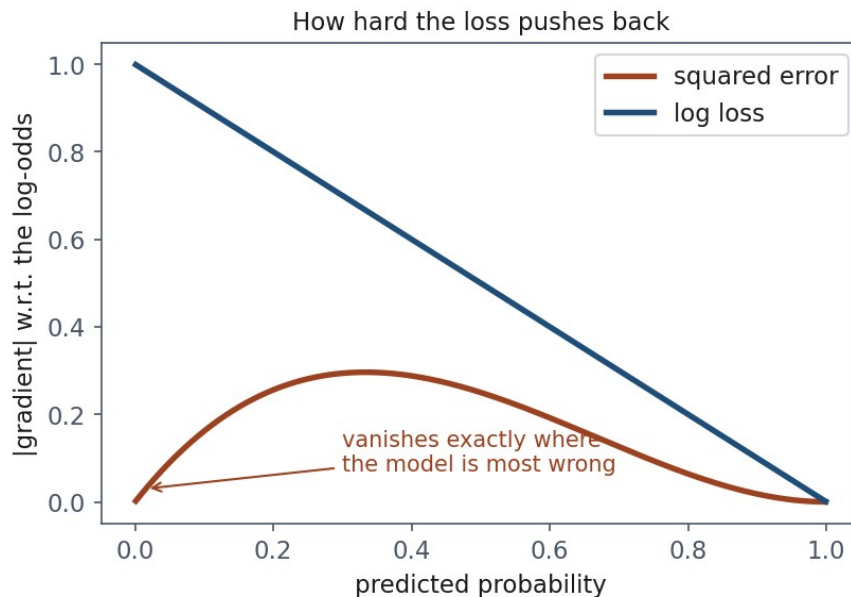
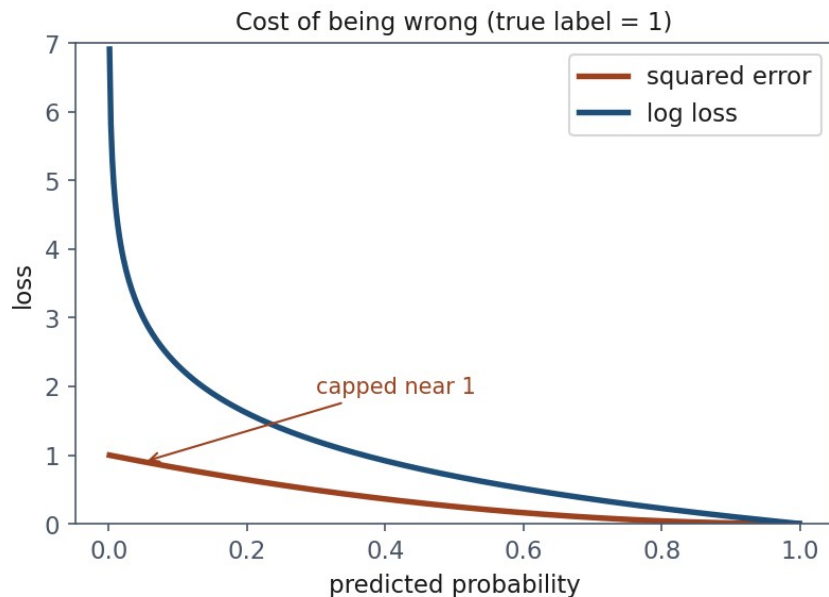
$$J(w, b) = -\frac{1}{m} \sum_{i=1}^m [y^{(i)} \log p^{(i)} + (1 - y^{(i)}) \log (1 - p^{(i)})]$$

The gradient is the one you already know

$$\frac{\partial J}{\partial w_j} = \frac{1}{m} \sum_{i=1}^m (\hat{y}^{(i)} - y^{(i)}) x_j^{(i)}$$



# So why not squared error?



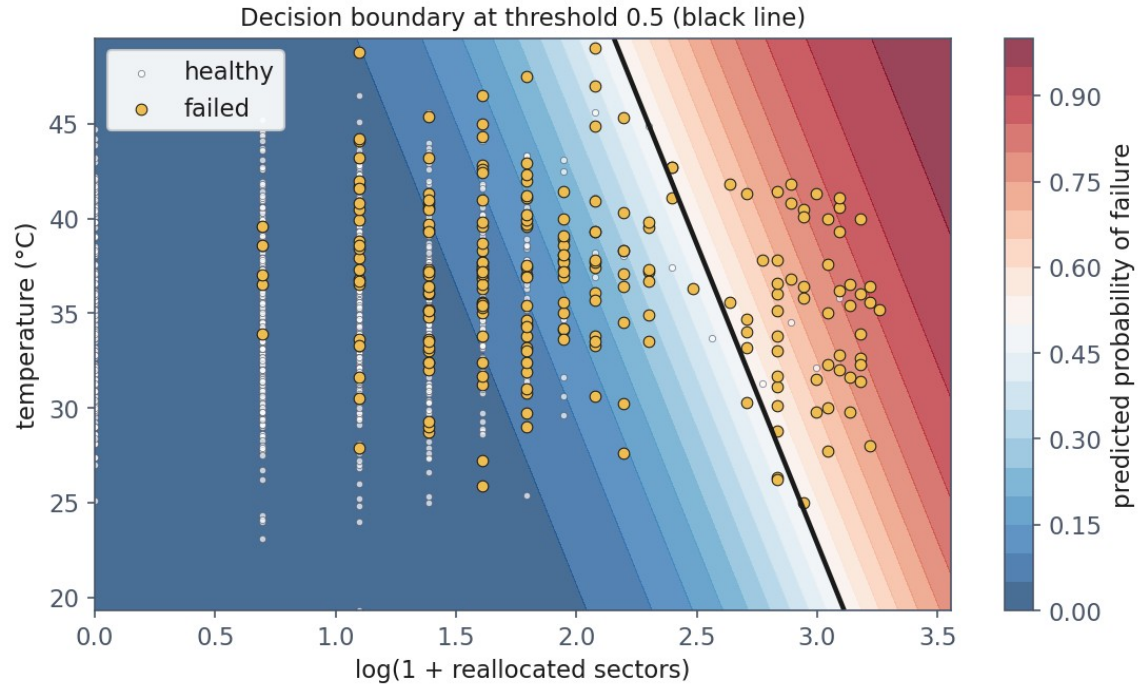
# And it is convex, which squared error is not

- Log loss is **convex** in the weights, so any local minimum is global
- Squared error through a sigmoid is not

# Notebook 1, live

- Fifteen lines of NumPy, then one line of scikit-learn
- Same answer

# The decision boundary



# Break

- Ten minutes

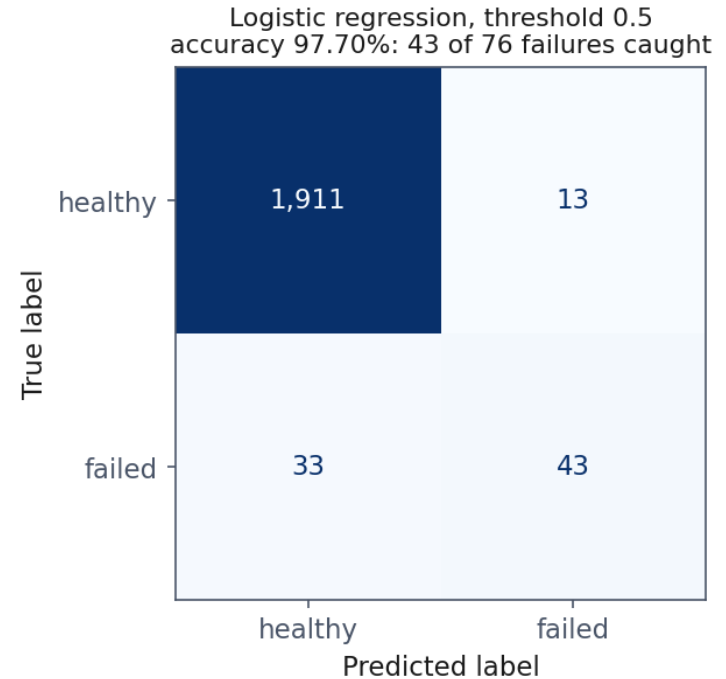
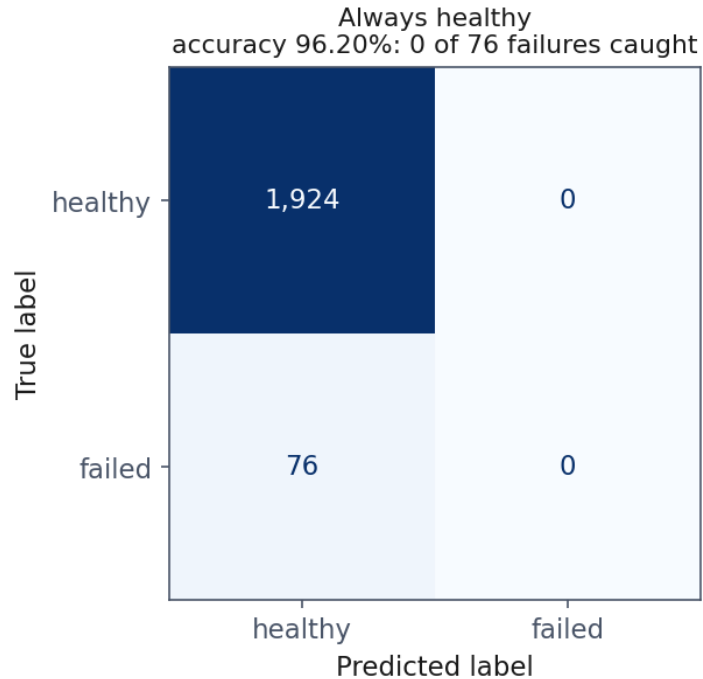
# Two models

| Model                              | Accuracy      | Failures caught |
|------------------------------------|---------------|-----------------|
| Predict “healthy” for everything   | <b>96.20%</b> | <b>0 of 76</b>  |
| Logistic regression, threshold 0.5 | 97.70%        | 43 of 76        |

# Why everyone makes this mistake once

- Accuracy is the fraction of predictions that were right
- On a **balanced** problem that is exactly what you want
- The interesting class is almost always the **rare** one

# Four numbers instead of one





# The four outcomes, and what they cost

- **False positive:** we replace a healthy drive. A technician's hour, €140
- **False negative:** a drive dies in service. Callout, rebuild, risk, €2,600

# Precision and recall: look at the denominator

Precision =  $TP / (TP + FP)$   
“of the alarms we raised”

|                  | predicted healthy | predicted failing |
|------------------|-------------------|-------------------|
| actually healthy | TN<br>1911        | <b>FP</b><br>13   |
| actually failed  | FN<br>33          | <b>TP</b><br>43   |

Recall =  $TP / (TP + FN)$   
“of the drives that failed”

|                  | predicted healthy | predicted failing |
|------------------|-------------------|-------------------|
| actually healthy | TN<br>1911        | FP<br>13          |
| actually failed  | <b>FN</b><br>33   | <b>TP</b><br>43   |

# Our model, in those terms

- Precision **0.768**: of 56 drives flagged, 43 really failed
- Recall **0.566**: of 76 drives that failed, we found 43
- Specificity 0.993: of 1,924 healthy drives, we left 1,911 alone

# F1: why the harmonic mean

| Model            | Precision | Recall | Arithmetic | Harmonic     |
|------------------|-----------|--------|------------|--------------|
| Flag every drive | 0.038     | 1.000  | 0.519      | <b>0.073</b> |

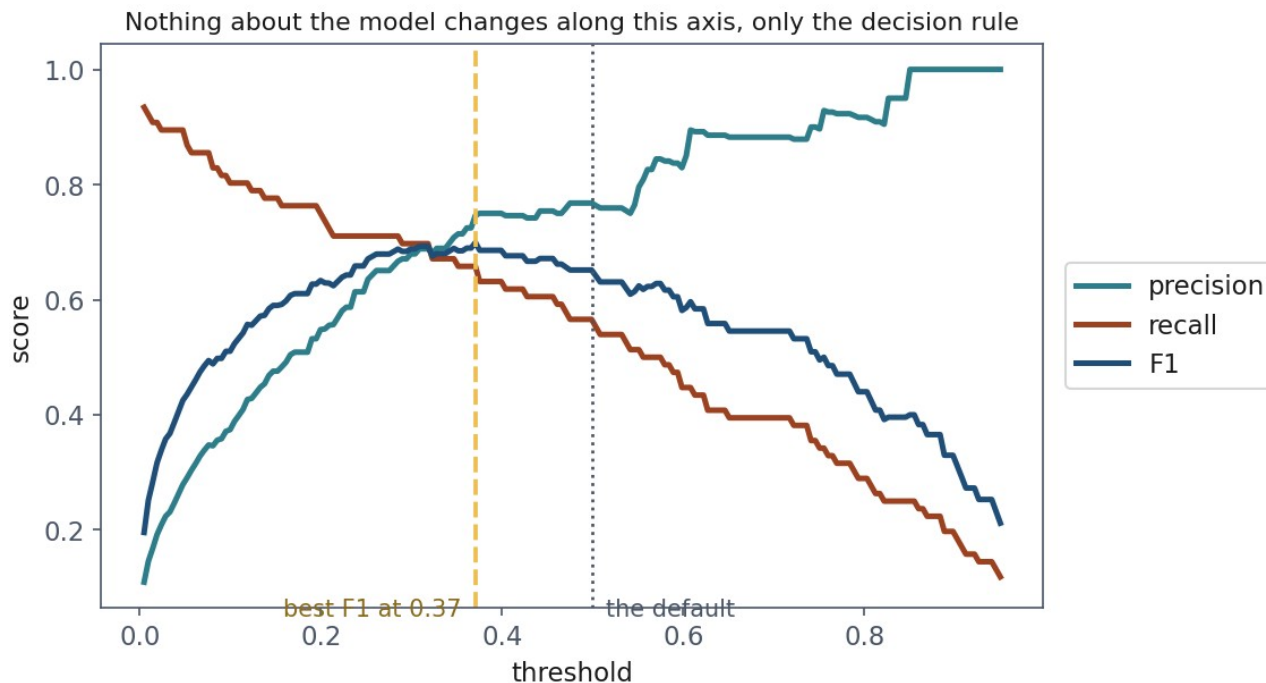
# Read the minority class, not the average

- weighted avg F1: **0.975** (dominated by the healthy class)
- macro avg F1: **0.820** (both classes get an equal say)

# Notebook 2, live

- The confusion matrix by hand, then from scikit-learn
- Then the threshold sweep

# The threshold is a choice nobody made

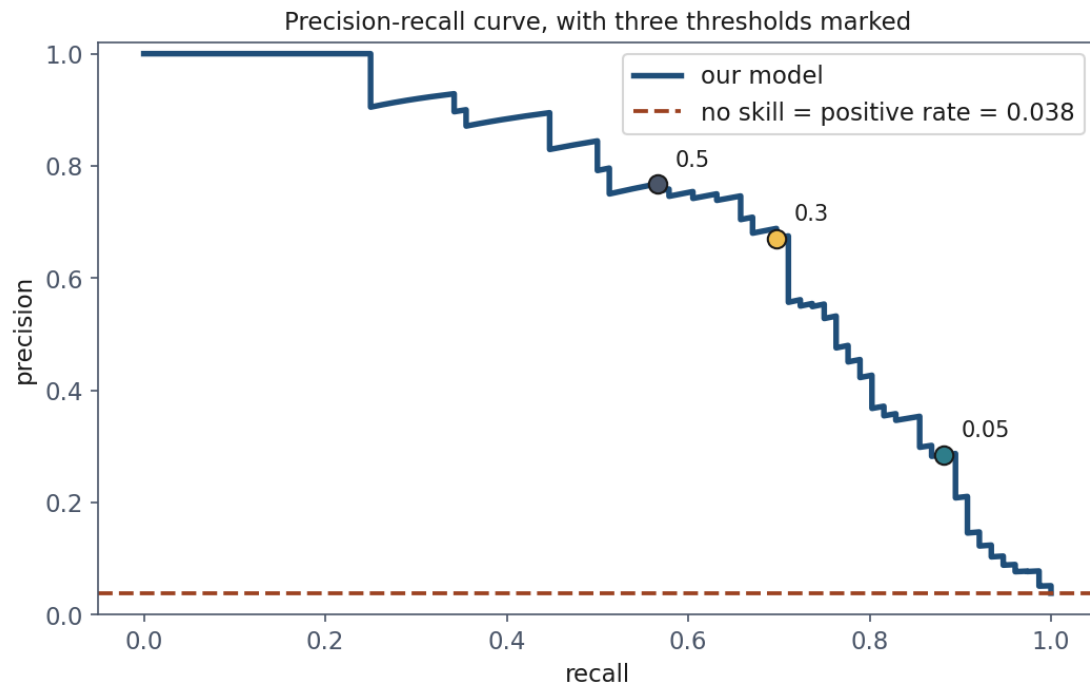


# What the sweep actually costs

| Threshold | TP | FP  | FN | Precision | Recall | Accuracy     |
|-----------|----|-----|----|-----------|--------|--------------|
| 0.02      | 69 | 284 | 7  | 0.195     | 0.908  | 0.855        |
| 0.10      | 61 | 102 | 15 | 0.374     | 0.803  | 0.942        |
| 0.30      | 53 | 25  | 23 | 0.679     | 0.697  | 0.976        |
| 0.50      | 43 | 13  | 33 | 0.768     | 0.566  | <b>0.977</b> |
| 0.90      | 14 | 0   | 62 | 1.000     | 0.184  | 0.969        |



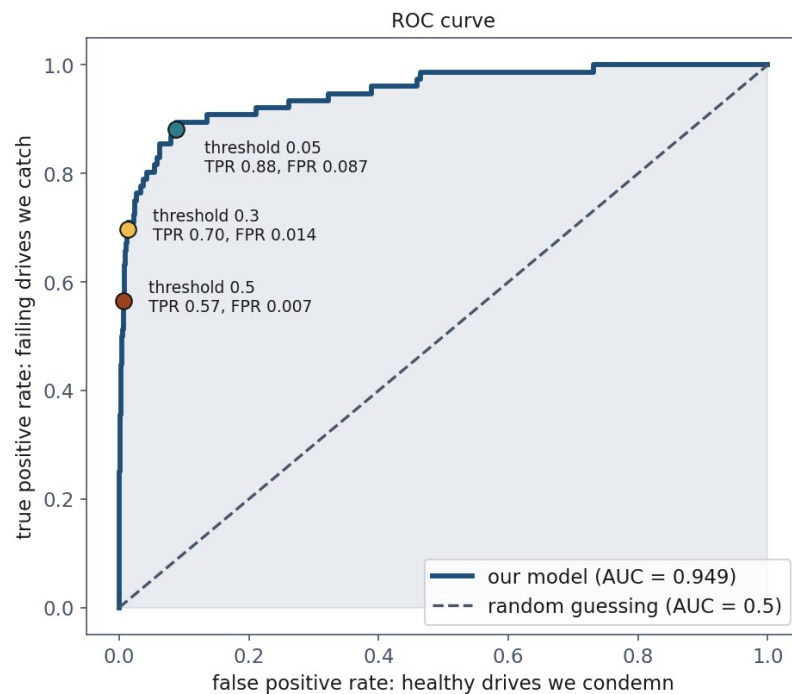
# The precision-recall curve



# Comparing models without choosing a threshold

- We keep evaluating at one threshold, but which?
- What if we score the whole **ranking** instead?

# The ROC curve



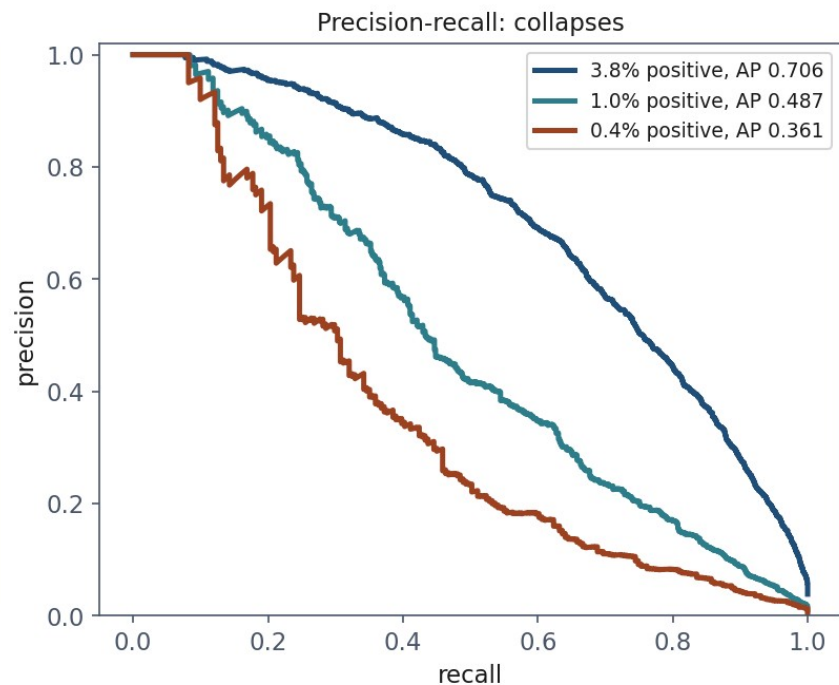
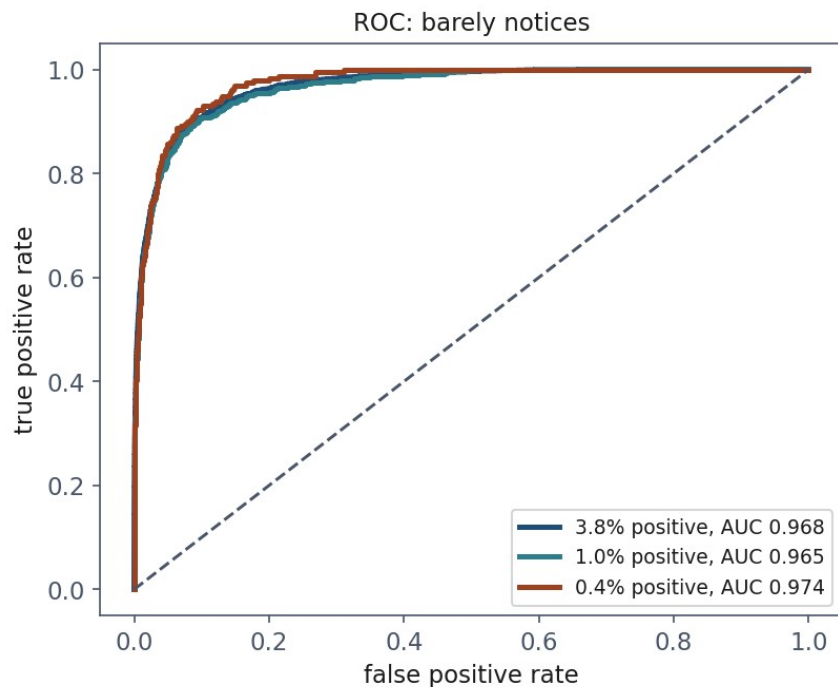
# What the area means

- **AUC = the probability that a random failing drive scores above a random healthy one**
- Ours: 0.949

# What AUC does not tell you

- Divide every predicted probability by 10
- AUC is **unchanged**; every decision changes

# Where ROC misleads



# Why the two disagree

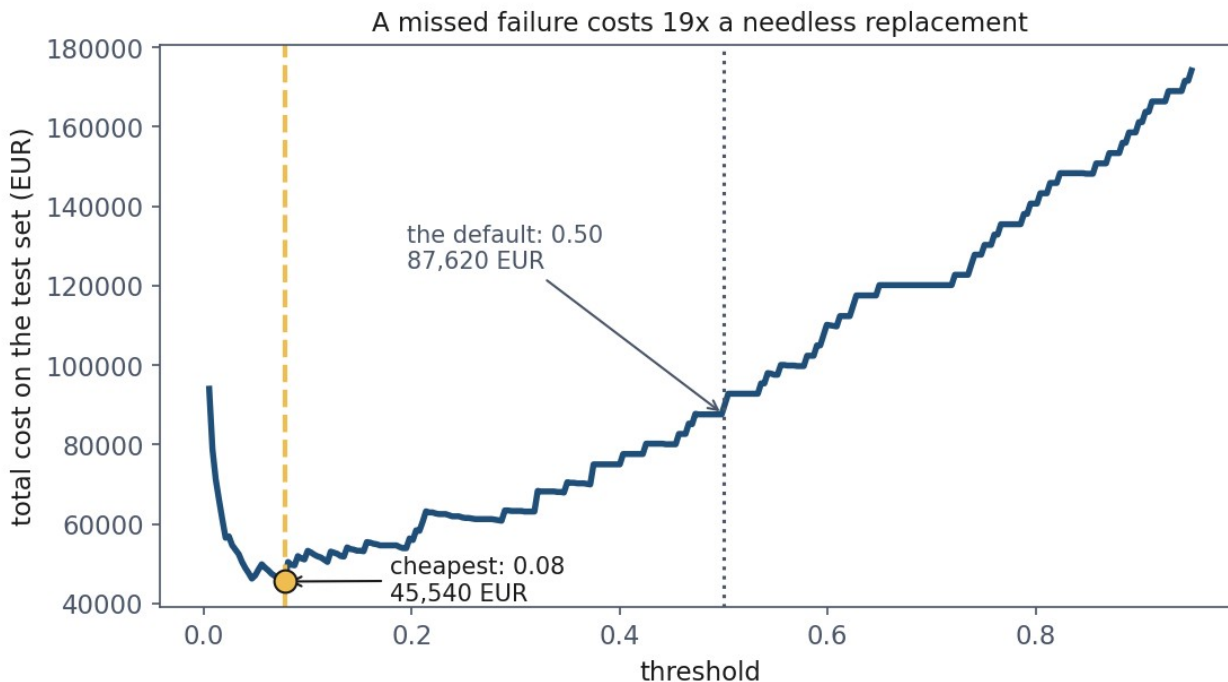
- FPR divides by **all healthy drives**: an enormous, growing denominator
- Precision divides by **the alarms raised**, where the false ones dominate

# The rule

- On an imbalanced problem, report the **precision-recall curve alongside AUC**
- Never instead of it, and never AUC alone



# Choosing the threshold from money



And it has a closed form

$$t^* = \frac{C_{FP}}{C_{FP} + C_{FN}} = \frac{140}{140 + 2600} = 0.051$$

# The caveat you must state

- We chose the threshold **by looking at the test set**
- That is precisely what lesson 1 forbade

# Class weights: the same lever, pulled earlier

| Model                  | Recall | Precision | AUC          | Cost   |
|------------------------|--------|-----------|--------------|--------|
| Plain, threshold 0.50  | 0.566  | 0.768     | 0.949        | 87,620 |
| Plain, threshold 0.08  | 0.855  | 0.349     | 0.949        | 45,540 |
| Balanced weights, 0.50 | 0.895  | 0.249     | <b>0.950</b> | 49,500 |

# Notebook 3, live

- ROC by hand, the 200,000-pair experiment, the cost curve
- Then the multiclass matrix

# More than two classes



# Macro against weighted, one last time

- macro avg F1: **0.707**
- weighted avg F1: **0.883**

# What to take away

- **96.20% accuracy, nothing caught.** Accuracy measures the majority class
- Precision is about the **alarms**, recall is about the **failures**
- The **threshold is a decision**, and it has a formula
- **AUC measures ranking** and hides imbalance



# Homework

- **Exercise 4**, due **Friday 23 October 2026, 09:00**
- Build a classifier, choose a threshold from a stated cost ratio, defend it